blink[well] — how the algorithm works

For stakeholders — what runs on the user's machine, and where Google's library fits.

WHAT IT DOES, IN ONE SENTENCE

Every ~10 minutes, blinkwell opens the user's camera for ~60 seconds, measures how their eyes are behaving (blink rate, eyelid closure, partial blinks), and compares that against *their own* normal — then suggests a break when fatigue builds. Everything happens on the device; no video ever leaves it.

THE PIPELINE, STEP BY STEP

1. **Short camera windows**
   The camera is opened only for a ~60-second check every ~10 minutes (~10% of the time) and fully released between checks — the webcam LED is an honest signal.

2. **Face landmarks — Google's MediaPipe Face Landmarker**
   Each frame is passed to Google's on-device model, which returns **478 3D facial landmarks**, **52 expression signals** ("blendshapes", including per-eye blink), and a **head-pose matrix** (head rotation/position). Runs locally in real time.

3. **Quality gates — only good frames count**
   Frames are discarded when there is no face, the head is turned too far, it's too dark, the frame rate dropped, or there's heavy motion. Look away mid-check and the clock pauses — only aligned, usable time is measured.

4. **Blink state machine**
   A small deterministic state machine tracks each eyelid's cycle — **open → closing → closed → reopening**. From complete and interrupted cycles it derives the fatigue signals below.

5. **Eight fatigue signals → one score**
   Signals are combined into a 0–100 fatigue score with a status and a suggested action (a 20-20-20 break, palming, or guided full-blink exercises).

6. **Personal baseline**
   The first 3 checks learn the user's *own* normal. After that, scores are judged against **their** baseline — not population averages — so the tool adapts to the person.

THE EIGHT SIGNALS IT MEASURES

Blink rate	Blinks per minute — drops when concentrating, spikes when strained.
Partial blinks	Share of blinks that don't fully close — a classic screen-fatigue sign.
PERCLOS	Percent of time the eyelids cover the eyes — the standard clinical drowsiness measure.
Blink duration	How long eyes stay closed per blink — longer as tiredness grows.

Reopen completeness	Whether eyes fully reopen — lagging lids signal fatigue.
Screen distance	Estimated from head geometry — leaning closer correlates with strain.
Usable check time	Share of the window with a well-aligned face — feeds confidence.
Confidence	How trustworthy this particular check is (lighting, alignment, duration).

WHAT GOOGLE'S LIBRARY PROVIDES — AND WHAT IT DOESN'T

We use **MediaPipe Face Landmarker**, Google's production face model. Its capabilities:

- **478 3D face landmarks** at real-time frame rates — the geometry we measure eyelid gaps, blink cycles and head distance from.
- **52 blendshape coefficients** — per-frame expression scores (e.g. `eyeBlinkLeft`, `eyeBlinkRight`) that corroborate the geometric signals.
- **Head-pose transform matrix** — head rotation/position, used for the alignment gates and distance estimate.
- **Fully on-device** — the model is a small file (~MB) embedded in our app/CLI; it needs no internet and runs on CPU (XNNPACK) or browser GPU (WebGL/wasm).

A note on provenance: Google ships no official Rust bindings, so our Rust CLI calls the engine through a small, pinned community crate (`mediapipe 0.1`) — a thin FFI shim with no ML code of its own. The engine it loads is the **official Google C-API build**, fetched once from Google's own PyPI wheel with a pinned SHA-256 checksum — the same binary Google's Python bindings run on.

Equally important — what it is **not** used for: no face recognition or identification, no images stored, no cloud calls for video. Frames are processed in memory and discarded immediately; only summary numbers (the signals above) are saved to the user's own history file.

> **One honest caveat:** the newest engine builds may send Google anonymous usage counters (which feature ran, how long it took). Camera frames and results are never sent. This is documented publicly on the product site.

WHY THE DESIGN IS THE WAY IT IS

- **Periodic, not continuous** — the camera is off ~90% of the time; the LED can't silently stay on, and it keeps the tool lightweight.
- **Personal, not population-average** — blink norms vary hugely between people; a per-user baseline is what makes the score meaningful.
- **Explainable signals** — every score decomposes into measurable eye behaviors, so the product can show *why* it's suggesting a break, not just *that* it is.